

Helix Thready — Configuration & Environment-Variable Reference

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Rev	Date	Author	Change
1	2026-07-21	swarm (user-guides)	Complete <code>.env</code> reference for the zero version
2	2026-07-22	swarm (user-guides, Pass 3)	Depth pass: corrected messenger env names to the VERIFIED Herald <code>HERALD_MTPROTO_*</code> / <code>HERALD_TGRAM_*</code> names read from <code>vasic-digital/herald/.env.example</code> ; added VERIFIED VisionEngine (<code>HELIX_VISION_*</code> , <code>HELIX_OLLAMA_*</code> , <code>HELIX_LLAMACPP_RPC_*</code>), LLMProvider (Cerebras/Fireworks/HuggingFace/Replicate/SambaNova/SiliconFlow) and additional <code>CONTAINERS_REMOTE_*</code> variables; added Appendix A — Master environment-variable index (name · purpose · default · scope · example); added Appendix B — worked <code>.env</code> examples per environment ; split the \$1 diagram explanation into multi-paragraph form; closed <code>[OPEN: cfg-messenger]</code> .

This is the **complete, documented environment-variable reference** for Helix Thready, mandated by the original request: “All environment variables that system supports *MUST* be properly documented in separate dedicated document(s).” Every variable the system reads is listed here with type, default, scope, provenance, and whether it is **VERIFIED** (a real in-house module variable read at source) or an **ASSUMPTION** / `[DEFAULT – adjustable]` (a Thready-specific default proposed here).

Naming convention (VERIFIED pattern, Thready names are ASSUMPTION). In-house modules use a module prefix: `HELIX_VISION_*` / `HELIX_OLLAMA_*` / `HELIX_LLAMACPP_RPC_*` (VisionEngine — VERIFIED in `helix_track/vision_engine/.env.example`), `HERALD_MTPROTO_*` / `HERALD_TGRAM_*` / `HERALD_MAX_*` (messengers — VERIFIED in `vasic-digital/herald/.env.example` and `herald/docs/guides/messengers/`), `CONTAINERS_REMOTE_*` (deployment — VERIFIED in `helix_track/containers/.env.example`), and bare provider keys such as `ANTHROPIC_API_KEY` , `OPENAI_API_KEY` , `GOOGLE_API_KEY` , `DEEPSEEK_API_KEY` (VERIFIED in `vasic-digital/LLMProvider/.env.example`). Thready’s own service variables use the **THREADY_*** prefix. The `THREADY_*` names are this document’s proposal `[DEFAULT –`

adjustable] ; the reused `HELIX_*` / `HERALD_*` / `CONTAINERS_*` / provider names are the actual module variables Thready inherits and **must not rename** — the exact spellings below were read from each module’s committed `.env.example` at source, not guessed.

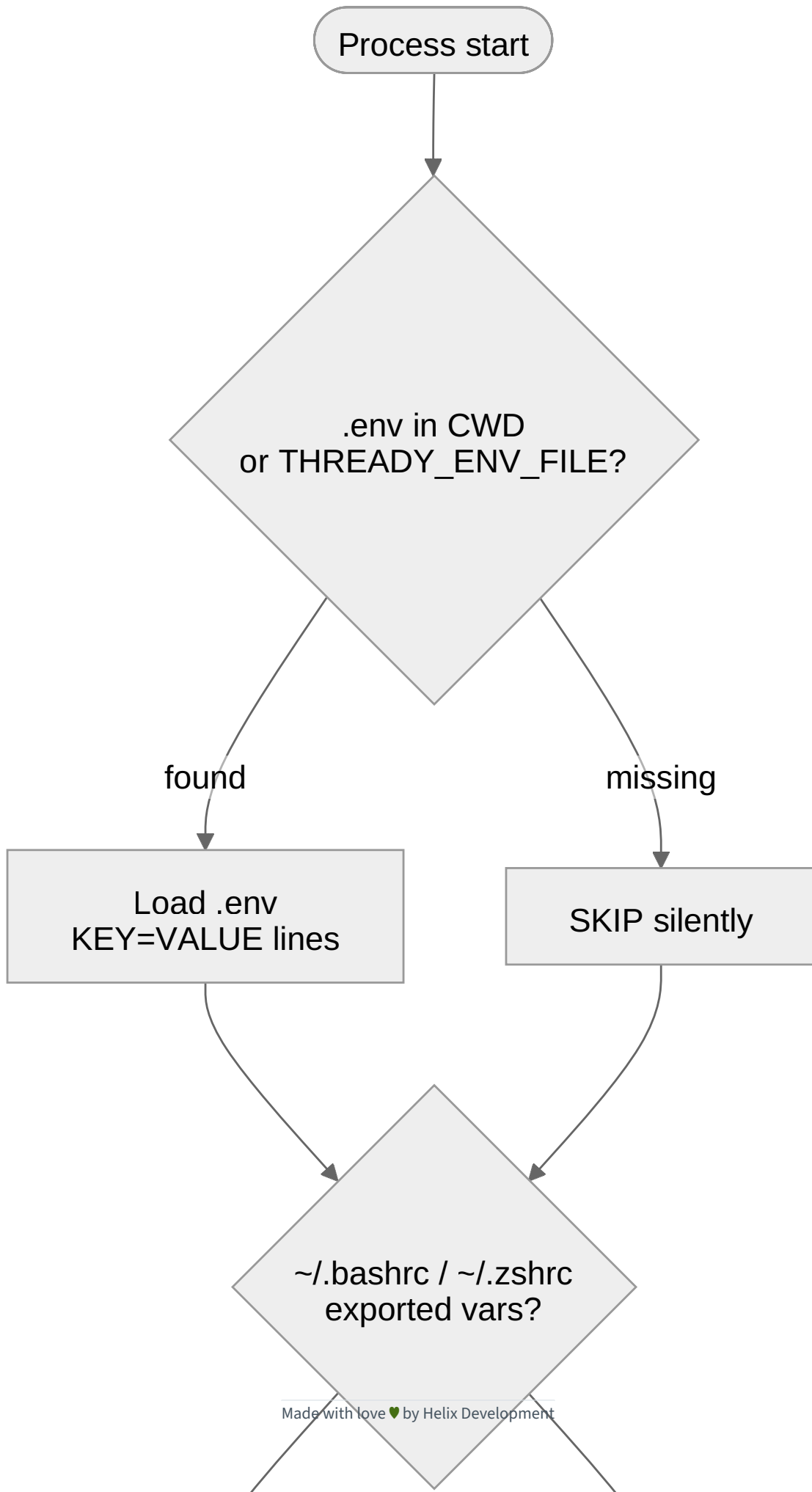
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1. How configuration is resolved

Per the original request, configuration comes from “*env. variables in .env file or obtaining them from .bashrc or .zshrc from host home directory*”, and credentials additionally from `$HOME/api_keys.sh` (Appendix B of the final request) and the private repo. Missing sources are **skipped silently** and never logged

`[CONSTITUTION §11.4.10]`.



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/config-resolution.mmd`.

Explanation (for readers/models that cannot see the diagram). At start-up each Thready service resolves configuration through an ordered chain of four optional sources, walking them top to bottom. Each source is skipped without error if it is absent — a missing `.env`, a bare shell, no `api_keys.sh`, and no secrets mount are all valid states that never abort the boot and never emit a log line about the absence. This is deliberate: the resolver is designed so the same binary boots identically on a developer laptop with a single `.env` and on a hardened production host where every secret arrives from a mounted vault, with no code path that treats “source not present” as an error.

The first source is a `.env` file in the working directory (or the explicit path in `THREADY_ENV_FILE`), parsed as simple `KEY=VALUE` lines. The second source is the **process environment** — any variable already exported by the operator’s `~/.bashrc` / `~/.zshrc`, a Kubernetes ConfigMap, or a CI secret store. The third source is `$HOME/api_keys.sh`, sourced for secrets (LLM provider keys, messenger tokens) — the operator’s shared key vault described in the final request’s Appendix B. The fourth source, used in production, is the **private-repo secrets mount**, which contributes sealed values that never touch the public tree.

The four sources are then **merged with a fixed precedence**: a value already present in the process environment wins over the same key in `.env`, which in turn wins over the compiled default. This is the identical 12-factor order the sibling Herald module documents as VERIFIED (`herald` spec V3 §3.3: explicit flag > shell export > `.env` fallback > compiled default), so an operator can keep team defaults in a committed-shaped `.env.example`-derived `.env` while overriding any single value per host via a shell export (`THREADY_LOG_LEVEL=debug ./thready serve`) — the file never clobbers the export.

Finally the service **validates** that every *required* variable for its role is present. If any are missing it **fails loudly**, printing the missing keys by name, and refuses to start half-configured; if a blocked value is present (for example `HELIX_EMBEDDING_PROVIDER=hash` in a search context, §8.1) it aborts with a fix-it message rather than warning-and-continuing. At no point are secret *values* printed — `*_PASSWORD` / `*_TOKEN` / `*_API_KEY` / `*_SECRET` / DSNs are redacted in every sink. This “SKIP-if-missing, fail-loud-if-required, never-log-secrets” behaviour is the Constitution’s secrets-hygiene rule `[CONSTITUTION §11.4.10]` applied uniformly across every Thready service.

2. The `.env` file and secret hygiene

- The `.env` file and any `secrets` / `api_keys.sh` **must be gitignored** and never committed to a public repo `[CONSTITUTION §11.4.10]`. Thready ships a committed `.env.example` template (placeholders only) and you copy it to `.env`.
- File permissions: `chmod 600 .env` and `chmod 700` on secret directories (VERIFIED requirement, final request §14.4 / Q39).
- Secrets are **runtime-load-only**; if a key leaks, rotate it and re-run the leak-audit hook.

- Sensitive values (`*_PASSWORD` , `*_TOKEN` , `*_API_KEY` , `*_SECRET` , DSNs) are **redacted in all logs and in the config-dump endpoint**. `[GAP: 7 / security]` The searchable-but-sealed representation for “encrypted yet semantically searchable credentials” is a `[BUILD-NEW]` item on `security/pkg/securestorage` ; until it lands, credential *content* extracted from posts is stored encrypted but not yet semantically indexed (see `end-user-manual.md`).

3. Quick-start minimal `.env`

The smallest `.env` that boots a **local development** stack (SQLite + in-process bus + local llama.cpp embeddings). Copy, then fill the messenger credentials to actually read a channel.

```

# — Core —————
THREADY_ENV=development
THREADY_HTTP_ADDR=0.0.0.0:8443
THREADY_LOG_LEVEL=info

# — Relational DB (dev = SQLite, cgo-free) —————
THREADY_DB_DRIVER=sqlite
THREADY_DB_DSN=file:./data/thready.db?_pragma=busy_timeout(5000)

# — Vector DB (dev = pgvector on a local Postgres) ———
THREADY_VECTOR_BACKEND=pgvector
THREADY_VECTOR_DSN=postgres://thready:thready@localhost:5432/thready?sslmode=disable
THREADY_EMBEDDING_DIM=1024

# — Embeddings: MUST be a real semantic provider ———
# [GAP: 1] Do NOT leave HelixLLM on its default hash embedder – it returns garbage relevance.
HELIX_EMBEDDING_PROVIDER=llama
THREADY_EMBEDDING_BASE_URL=http://localhost:8080/v1
THREADY_EMBEDDING_MODEL=jina-embeddings-v2-base-code

# — Event bus (dev = in-process) —————
THREADY_EVENTBUS_BACKEND=inprocess

# — Auth (dev secret; use RS256 keypair in prod) ———
THREADY_JWT_SIGNING_ALG=HS256
THREADY_JWT_SECRET=change-me-32-bytes-minimum-dev-only

# — Telegram (VERIFIED Herald MTProto user-client names) —
# Fill from api_keys.sh in real use. These read the FULL thread history.
HERALD_MTPROTO_APP_ID=
HERALD_MTPROTO_APP_HASH=
HERALD_MTPROTO_PHONE=
# HERALD_MTPROTO_PASSWORD=          # only if the account has cloud 2FA
# HERALD_MTPROTO_SESSION_FILE=~/.config/herald/mtproto.session
# — Telegram Bot API (status replies; cannot backfill history) —
# HERALD_TGRAM_BOT_TOKEN=
# HERALD_TGRAM_CHAT_ID=

```

4. Core runtime & server

Variable	Type	Default	Provenance	Notes
<code>THREADY_ENV</code>	enum <code>development</code> <code>staging</code> <code>production</code>	<code>development</code>	ASSUMPTION	Selects env-specific defaults; drives log format & safety rails.
<code>THREADY_ENV_FILE</code>	path	<code>./.env</code>	ASSUMPTION	Explicit <code>./.env</code> location (overrides CWD lookup).
<code>THREADY_HTTP_ADDR</code>	host:port	<code>0.0.0.0:8443</code>	ASSUMPTION	REST <code>/v1</code> + WS/SSE bind address.
<code>THREADY_HTTP3_ENABLED</code>	bool	<code>true</code>	<code>[IN-HOUSE: http3]</code>	Enables HTTP/3 (QUIC) via <code>basic-digital/http3</code> ; falls back to HTTP/2.
<code>THREADY_HTTP_COMPRESSION</code>	csv	<code>br,gzip</code>	<code>[DEFAULT — adjustable]</code>	Response compression (Brotli/gzip).
<code>THREADY_LOG_LEVEL</code>	enum <code>debug</code> <code>info</code> <code>warn</code> <code>error</code>	<code>info</code>	<code>[IN-HOUSE: observability]</code>	logrus level.
<code>THREADY_LOG_FORMAT</code>	enum <code>json</code> <code>text</code>	<code>json</code> (<code>text</code> in dev)	<code>[IN-HOUSE: observability]</code>	Structured logging.
<code>THREADY_REQUEST_TIMEOUT</code>	duration	<code>30s</code>	<code>[DEFAULT — adjustable]</code>	Per-request server timeout.
<code>THREADY_RATE_LIMIT_RPS</code>	int	<code>100</code>	<code>[IN-HOUSE: ratelimiter]</code>	Per-identity request rate cap.
<code>THREADY_CORS_ORIGINS</code>	csv	<code>`(none)   [IN-HOUSE: middleware]   Allowed browser origins (portal domains).     THREADY_PORT_PREFIX`</code>	Deterministic dynamic-port base for the container stack.	

Variable	Type	Default	Provenance	Notes
		<code>int (≤65535 budget) unset [IN-HOUSE: port_prefix]`</code>		

5. Environments, domains & TLS

Three fully separated environments behind subdomains on one Hetzner host (final request §8.2).

Variable	Type	Default	Provenance	Notes
THREADY_PUBLIC_DOMAIN	host	thready.hxd3v.com	[OPERATOR]	Base domain. Dev/staging derive dev. / sta. prefixes.
THREADY_PUBLIC_BASE_URL	url	https://thready.hxd3v.com	ASSUMPTION	Absolute base for links in generated content & emails.
LETS_ENCRYPT_EMAIL	email	unset	[IN-HOUSE: lets_encrypt]	ACME account email.
LETS_ENCRYPT_CHALLENGE	enum http-01 dns-01	http-01	[IN-HOUSE: lets_encrypt]	ACME challenge type (acme.sh).
THREADY_TLS_MIN_VERSION	enum	1.3	[DEFAULT – adjustable]	Minimum TLS version (§Q38).
CONTAINERS_REMOTE_ENABLED	bool	false	[IN-HOUSE: containers] (VERIFIED)	Enable remote container distribution.
CONTAINERS_REMOTE_DEFAULT_SSH_USER	string	deploy	[IN-HOUSE: containers] (VERIFIED)	SSH user for deploy targets (Thready uses thready).
CONTAINERS_REMOTE_DEFAULT_RUNTIME	enum docker podman	podman	[CONSTITUTION §11.4.76]	Rootless Podman only for Thready.
CONTAINERS_REMOTE_PORT_RANGE_START / _END	int	20000 / 30000	[IN-HOUSE: containers] (VERIFIED)	Tunnel/port range.
CONTAINERS_REMOTE_DEFAULT_SSH_KEY	path	unset	[IN-HOUSE: containers] (VERIFIED)	Private key for deploy-target SSH (else the agent).
CONTAINERS_REMOTE_CONNECT_TIMEOUT	duration	10s	[IN-HOUSE: containers] (VERIFIED)	SSH connect timeout to a deploy target.
CONTAINERS_REMOTE_COMMAND_TIMEOUT	duration	5m	[IN-HOUSE: containers] (VERIFIED)	Per-remote-command timeout.
CONTAINERS_REMOTE_SSH_CONTROL_MASTER	bool	true	[IN-HOUSE: containers] (VERIFIED)	Re-use one SSH control connection (ControlMaster).
	duration	60s		

Variable	Type	Default	Provenance	Notes
<code>CONTAINERS_REMOTE_SSH_CONTROL_PERSIST</code>			<code>[IN-HOUSE: containers]</code> (VERIFIED)	How long the multiplexed SSH master lingers.
<code>CONTAINERS_REMOTE_SSH_MAX_CONNECTIONS</code>	int	<code>10</code>	<code>[IN-HOUSE: containers]</code> (VERIFIED)	Concurrent SSH connections per target.
<code>CONTAINERS_REMOTE_SSH_SCHEDULER</code>	enum <code>roundrobin</code> <code>\ </code> <code>leastload</code> <code>d</code>	<code>roundrobin</code>	<code>[IN-HOUSE: containers]</code> (VERIFIED)	Placement across multiple remote hosts.
<code>CONTAINERS_REMOTE_VOLUME_TYPE</code>	enum <code>bind\ </code> <code>volume</code>	<code>volume</code>	<code>[IN-HOUSE: containers]</code> (VERIFIED)	Default mount type for remote deploys.

6. Relational database

`digital.vasic.database` selects the backend by `Driver` (VERIFIED: SQLite dev / Postgres prod, migrations via `pkg/migration.Runner`).

Variable	Type	Default	Provenance	Notes
<code>THREADY_DB_DRIVER</code>	enum <code>sqlite</code> <code>postgres</code>	<code>sqlite</code> (dev) / <code>postgres</code> (prod)	[IN-HOUSE: database]	Maps to <code>Config.Driver</code> .
<code>THREADY_DB_DSN</code>	dsn	see quick-start	[IN-HOUSE: database]	SQLite: <code>file:...;</code> Postgres: <code>postgres://user:pass@host:5432/db</code> .
<code>THREADY_DB_MAX_OPEN_CONNS</code>	int	<code>32</code>	[DEFAULT — adjustable]	pgx pool size (tune for Large scale).
<code>THREADY_DB_MAX_IDLE_CONNS</code>	int	<code>8</code>	[DEFAULT — adjustable]	Idle pool.
<code>THREADY_DB_CONN_MAX_LIFETIME</code>	duration	<code>30m</code>	[DEFAULT — adjustable]	Connection recycle.
<code>THREADY_DB_MIGRATE_ON_BOOT</code>	bool	<code>true</code> (dev) / <code>false</code> (prod)	[IN-HOUSE: database]	Auto-run <code>migration.Runner.Apply</code> at start (prod runs migrations via deploy step).
<code>THREADY_DB_PARTITIONING</code>	bool	<code>true</code> (prod)	[GAP: 3.2]	Time-partitioned <code>posts</code> for 10k+/day. Partitioning helpers are a <code>P1</code> improvement on <code>database</code> — until merged this is best-effort.

7. Datastores

7.1 Vector DB (semantic search) [GAP: 8]

Only the **pgvector** backend is production-wired (VERIFIED); Qdrant/Pinecone/Milvus adapters exist behind the same `VectorStore` interface but are **unverified end-to-end** — a config-only swap is planned, not guaranteed.

Variable	Type	Default	Provenance	Notes
<code>THREADY_VECTOR_BACKEND</code>	enum <code>pgvector</code> <code>qdrant</code> <code>pinecone</code> <code>milvus</code>	<code>pgvector</code>	[IN-HOUSE: vectordb]	Only <code>pgvector</code> is VERIFIED. Others [OPEN].
<code>THREADY_VECTOR_DSN</code>	dsn	shares Postgres	[IN-HOUSE: vectordb]	pgvector co-locates in the relational Postgres.
<code>THREADY_VECTOR_METRIC</code>	enum <code>cosine</code> <code>l2</code> <code>ip</code>	<code>cosine</code>	[IN-HOUSE: vectordb]	Cosine <code><=></code> per decision matrix.
<code>THREADY_EMBEDDING_DIM</code>	int	<code>1024</code>	[GAP: 1]	Must match the embedding model's true dimension. RAG path historically hardcoded 768 — set this explicitly.
<code>THREADY_VECTOR_INDEX</code>	enum <code>hnsw</code> <code>ivfflat</code>	<code>hnsw</code>	[DEFAULT – adjustable]	ANN index for the < 500 ms search SLO.
<code>THREADY_QDRANT_URL</code>	url	unset	[GAP: 8]	Only used if backend= <code>qdrant</code> ; hardening tracked as [OPEN].

7.2 Cache

Variable	Type	Default	Provenance	Notes
<code>THREADY_CACHE_BACKEND</code>	enum <code>memory</code> <code>redis</code> <code>postgres</code>	<code>memory</code> (dev) / <code>redis</code> (prod)	<code>[IN-HOUSE: cache]</code>	L1/L2 tiers supported.
<code>THREADY_CACHE_REDIS_URL</code>	url	unset	<code>[IN-HOUSE: cache]</code>	<code>redis://host:6379/0</code> .
<code>THREADY_CACHE_TTL</code>	duration	<code>10m</code>	<code>[DEFAULT — adjustable]</code>	Default entry TTL.

7.3 Object storage (assets)

`digital.vasic.storage` (MinIO/S3) is the asset tier for the 50 TB+ scale.

Variable	Type	Default	Provenance	Notes
<code>THREADY_STORAGE_BACKEND</code>	enum <code>filesystem</code> <code>minio</code> <code>s3</code>	<code>filesystem</code> (dev) / <code>minio</code> (prod)	<code>[IN-HOUSE: storage]</code>	
<code>THREADY_STORAGE_ENDPOINT</code>	url	unset	<code>[IN-HOUSE: storage]</code>	MinIO endpoint.
<code>THREADY_STORAGE_BUCKET</code>	string	<code>thready-assets</code>	ASSUMPTION	Object bucket.
<code>THREADY_STORAGE_ACCESS_KEY</code>	secret	unset	<code>[IN-HOUSE: storage]</code>	Redacted in logs.
<code>THREADY_STORAGE_SECRET_KEY</code>	secret	unset	<code>[IN-HOUSE: storage]</code>	Redacted in logs.
<code>THREADY_STORAGE_SIGNED_URL_TTL</code>	duration	<code>15m</code>	<code>[GAP: 3.2]</code>	MinIO signed-URL parity with CloudFront signing is a <code>P1</code> verification item.

8. Embeddings, LLM & Vision

8.1 Embeddings `[GAP: 1]` (P0 trap — read this)

VERIFIED danger zone. HelixLLM's *default* local embedder is a non-semantic `HashEmbedder` stub that emits deterministic pseudo-vectors with only a startup WARNING; any semantic search built on the default

silently returns garbage. **You must set** `HELIX_EMBEDDING_PROVIDER=llama` (real llama.cpp `/embedding`) for every semantic workload. Thready is configured to **fail loudly**, not warn, if the hash embedder is selected in a search/RAG context.

Variable	Type	Default	Provenance	Notes
<code>HELIX_EMBEDDING_PROVIDER</code>	enum <code>llama\ openai\ hash</code>	<code>llama</code> (enforced)	<code>[IN-HOUSE: HelixLLM]</code> (VERIFIED)	<code>hash</code> is blocked for search; setting it aborts start-up.
<code>THREADY_EMBEDDING_BASE_URL</code>	url	<code>http://localhost:8080/v1</code>	<code>[IN-HOUSE: embeddings]</code>	OpenAI-compatible endpoint (HelixLLM <code>/v1/embeddings</code>).
<code>THREADY_EMBEDDING_MODEL</code>	string	<code>jina-embeddings-v2-base-code</code>	<code>[RESEARCH]</code>	Code-tuned; alt <code>voyage-code-3</code> .
<code>THREADY_EMBEDDING_API_KEY</code>	secret	unset	<code>[IN-HOUSE: embeddings]</code>	Only if the endpoint requires it.

8.2 LLM providers

Bare provider-key names are **VERIFIED** from `llm_provider/.env.example`. Set only the ones you use; missing keys are skipped.

Variable	Type	Provenance	Notes
<code>HELIX_LLM_BASE_URL</code>	url	<code>[IN-HOUSE: HelixLLM]</code>	Local llama.cpp server (OpenAI+Anthropic APIs).
<code>HELIX_LLM_MODEL</code>	string	<code>[IN-HOUSE]</code>	e.g. <code>Llama-3.1-70B-Instruct-Q4_K_M</code> (research/reasoning).
<code>HELIX_LLM_CODE_MODEL</code>	string	<code>[IN-HOUSE]</code>	e.g. <code>Qwen2.5-Coder</code> .
<code>ANTHROPIC_API_KEY</code>	secret	VERIFIED	Cloud fallback <code>claude-sonnet-4</code> .
<code>OPENAI_API_KEY</code>	secret	VERIFIED	Fallback / vision <code>gpt-4o</code> .
<code>GOOGLE_API_KEY</code>	secret	VERIFIED	Fallback <code>gemini-2.5-pro</code> .
<code>DEEPSEEK_API_KEY</code>	secret	VERIFIED	Fallback <code>deepseek-v3</code> .
<code>GROQ_API_KEY</code> , <code>MISTRAL_API_KEY</code> , <code>QWEN_API_KEY</code> , <code>OPENROUTER_API_KEY</code> , <code>COHERE_API_KEY</code> , <code>TOGETHER_API_KEY</code> , <code>XAI_API_KEY</code> , <code>PERPLEXITY_API_KEY</code> , <code>NVIDIA_API_KEY</code>	secret	VERIFIED	Additional <code>LLMProvider</code> adapters; set as needed.
<code>CEREBRAS_API_KEY</code> , <code>FIREWORKS_API_KEY</code> , <code>HUGGINGFACE_API_KEY</code> , <code>SAMBANOVA_API_KEY</code> , <code>SILICONFLOW_API_KEY</code>	secret	VERIFIED (<code>LLMProvider/.env.example</code>)	Further VERIFIED provider adapters.
<code>REPLICATE_API_TOKEN</code>	secret	VERIFIED (<code>LLMProvider/.env.example</code>)	Replicate uses a <code>*_TOKEN</code> , not <code>*_API_KEY</code> , spelling — do not rename.
<code>THREADY_LLM_MAX_RETRIES</code>	int (<code>5</code>)	<code>[DEFAULT — adjustable]</code>	Provider retry ceiling.
<code>THREADY_LLM_CIRCUIT_BREAKER</code>	bool (<code>true</code>)	<code>[IN-HOUSE: LLMProvider]</code>	Circuit breaker for flapping providers.

8.3 Vision & OCR [GAP: 2]

`HELIX_VISION_*` are **VERIFIED** in `vision_engine/.env.example`. VisionEngine has **no OCR engine** (P0 gap) — OCR is a `[BUILD-NEW]` Tesseract/PaddleOCR adapter. Until it lands, `#Comic` / `#Screenshot` OCR falls back to LLM-vision transcription only (lower fidelity, non-deterministic).

Variable	Type	Default	Provenance	Notes
<code>HELIX_VISION_PROVIDER</code>	enum <code>auto\ </code> <code>openai\ </code> <code> </code> <code>anthropic\ </code> <code>gemin\ </code> <code>qwen\ </code> <code>astica\ </code> <code> </code> <code>nvidia\ </code> <code>kimi\ </code> <code>stepfun\ </code> <code>\ </code> <code>ollama</code>	<code>auto</code>	<code>[IN-HOUSE: VisionEngine]</code> (VERIFIED)	Probes configured providers. Enum widened to the VERIFIED set in <code>vision_engine/.env.example</code> .
<code>HELIX_VISION_TIMEOUT</code>	int (sec)	<code>60</code>	VERIFIED	Vision call timeout.
<code>HELIX_VISION_MAX_IMAGE_SIZE</code>	int (px)	<code>4096</code>	VERIFIED	Downscale ceiling.
<code>HELIX_VISION_OPENCV_ENABLED</code>	bool	<code>true</code>	VERIFIED (<code>vision_engine/.env.example</code>)	Enable the OpenCV pre-analysis pass (SSIM/near-duplicate detection).
<code>HELIX_VISION_SSIM_THRESHOLD</code>	float <code>0..1</code>	<code>0.95</code>	VERIFIED	Structural-similarity cutoff for treating two frames/images as duplicates.
<code>HELIX_VISION_HOSTS</code>	csv	unset	VERIFIED	Vision worker hosts to distribute calls across (e.g. <code>thinker.local</code> , <code>amber.local</code>).
<code>HELIX_VISION_USER</code>	string	unset	VERIFIED	SSH user for the remote vision hosts.
<code>HELIX_OLLAMA_URL</code>	url	<code>http://localhost:1434</code>	VERIFIED	Local Ollama endpoint when <code>HELIX_VISION_PROVIDER=ollama</code> .
<code>HELIX_OLLAMA_MODEL</code>	string	<code>minicpm-v:8b</code>	VERIFIED	Local multimodal model for Ollama vision.
<code>HELIX_LLAMACP_RPC_ENABLED</code>	bool	<code>false</code>	VERIFIED	Use a llama.cpp RPC worker pool for local vision.
		unset	VERIFIED	

Variable	Type	Default	Provenance	Notes
<code>HELIX_LLAMACP_RPC_WORKERS</code>	csv host:port			RPC worker addresses (only if RPC enabled).
<code>HELIX_LLAMACP_RPC_MODEL</code>	path	<code>~/models/vision-model.gguf</code>	VERIFIED	GGUF model for the RPC vision path.
<code>ASTICA_API_KEY</code>	secret	unset	VERIFIED (<code>vision_engine/.env.example</code>)	API key for the <code>astica</code> vision provider.
<code>KIMI_API_KEY</code> , <code>STEPFUN_API_KEY</code>	secret	unset	VERIFIED	Keys for the <code>kimi</code> / <code>stepfun</code> vision providers.
<code>THREADY_OCR_PROVIDER</code>	enum <code>tesseract\ paddleocr\ none</code>	<code>none</code>	<code>[BUILD-NEW]</code>	Real OCR pending the adapter; <code>none</code> = LLM-vision only.
<code>THREADY_OCR_LANGS</code>	csv	<code>eng,rus</code>	<code>[DEFAULT — adjustable]</code>	Tesseract language packs. Add <code>srp</code> for Serbian-Cyrillic content — <code>sr-Cyrl</code> is a Full-priority UI locale (\$15), so deployments serving Serbian scans should set <code>eng,rus,srp</code> .

VERIFIED note on OpenCV vs OCR. VisionEngine’s OpenCV pass (`HELIX_VISION_OPENCV_ENABLED`, `HELIX_VISION_SSIM_THRESHOLD`) is real and does image-similarity / near-duplicate work — but it is **not** text recognition. The `[GAP: 2]` “no OCR engine” status stands: OpenCV de-duplicates frames, it does not transcribe `#Comic` / `#Screenshot` text. Do not mistake OpenCV-enabled for OCR-capable.

9. Messengers (Herald) `[GAP: 3]`

VERIFIED status (read from `vasic-digital/herald` source). Herald exposes **two distinct Telegram transports**, and Thready uses both for different jobs:

- The **MTPROTO user client** (`HERALD_MTPROTO_*`) signs in as a real Telegram **user account** and is the only transport that can **read full thread history / backfill** a channel. It exists in the code today but lives inside Herald’s `qaherald` QA harness and is being **promoted to a first-class channel** for Thready (`[BUILD-NEW] P0`). This is the transport Thready’s ingest depends on.
- The **Bot-API path** (`HERALD_TGRAM_*`) is a `@BotFather` bot used for **outbound status replies** and an optional inbound long-poll. It **cannot backfill channel history** — Telegram’s bot-privacy boundary

blocks a bot from reading other members' history — so it is a *reply/notify* transport, not a *reader* (VERIFIED: [herald/docs/research/telegram-bot-to-bot-constraint.md](#)).

Max is an empty stub. Only reserved env vars exist (`HERALD_MAX_BOT_TOKEN` , `HERALD_MAX_CHAT_ID`); Herald's own [docs/guides/messengers/MAX.md](#) marks Max **PLANNED (HRD-NNN)** with “(not yet implemented)”. The adapter (Bot API + Go port of the OneMe user-WebSocket) is `[BUILD-NEW]` P0. Do not expect Max reading before that lands. Sign-in flows: [installation.md](#) §7.

Telegram — MTPROTO user client (reads history):

Variable	Type	Default	Provenance	Notes
<code>HERALD_MTPROTO_APP_ID</code>	secret (int)	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED — <code>.env.example</code>)	<code>app_id</code> from my.telegram.org/apps (~5–8 digits).
<code>HERALD_MTPROTO_APP_HASH</code>	secret (32-hex)	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Paired <code>app_hash</code> from my.telegram.org/apps .
<code>HERALD_MTPROTO_PHONE</code>	string (E.164)	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Phone of the reading user account; first run triggers an SMS/app login code.
<code>HERALD_MTPROTO_PASSWORD</code>	secret	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Cloud 2FA password; only prompted if Telegram returns <code>SESSION_PASSWORD_NEEDED</code> .
<code>HERALD_MTPROTO_SESSION_FILE</code>	path	<code>~/.config/herald/mtproto.session</code>	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Persisted MTPROTO session (avoids re-login). Store on an encrypted path.

Telegram — Bot API (status replies / notify):

Variable	Type	Default	Provenance	Notes
<code>HERALD_TGRAM_BOT_TOKEN</code>	secret	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED)	<code>@BotFather</code> token (<code>NNNNNNNNNN:35-char</code>). Cannot backfill history.
<code>HERALD_TGRAM_CHAT_ID</code>	string	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Target chat: private DM positive, group negative, supergroup <code>-100<id></code> .
<code>HERALD_TGRAM_LIVE_INBOUND</code>	bool	<code>false</code>	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Enable the Bot-API inbound long-poll listener.
<code>HERALD_OPERATOR_IDS</code>	csv (int)	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED)	Telegram user IDs allowed to issue operator commands.

Max (reserved — PLANNED, not implemented) & Thready messenger orchestration:

Variable	Type	Default	Provenance	Notes
<code>HERALD_MAX_BOT_TOKEN</code>	secret	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED reserved)	Max Bot-API token — adapter PLANNED, not built (<code>herald/docs/guides/messengers/MAX.md</code>).
<code>HERALD_MAX_CHAT_ID</code>	string	unset	<code>[IN-HOUSE: herald]</code> (VERIFIED reserved)	Reserved; no code consumes it yet.
<code>THREASY_MESSENGER_SIGNIN_MODE</code>	enum <code>interactive</code> <code>noninteractive</code>	<code>interactive</code>	<code>[OPERATOR]</code>	Non-interactive reads all creds from env for headless deploy.
<code>THREASY_POLL_INTERVAL</code>	duration	<code>5m</code>	<code>[DEFAULT – adjustable]</code>	Scheduled wake frequency; event triggers supplement polling.
<code>THREASY_REPLY_ACCOUNT</code>	enum <code>robot</code> <code>user</code>	<code>robot</code>	<code>[DEFAULT – adjustable]</code>	Which account posts status replies (<code>robot</code> → Bot API, <code>user</code> → MTPROTO).

Migration note (from Rev 1). Earlier drafts used `HERALD_TELEGRAM_API_ID` / `_API_HASH` / `_PHONE` / `_2FA_PASSWORD` / `_SESSION_PATH`. Those names were an ASSUMPTION and are **wrong** — the VERIFIED Herald spellings are the `HERALD_MTPROTO_*` names above. If you copied an early `.env`, rename those five keys.

10. Event bus, background jobs & processing

Variable	Type	Default	Provenance	Notes
<code>THREADY_EVENTBUS_BACKEND</code>	enum <code>inprocess</code> <code>inprocess\</code> <code>nats</code>	<code>inprocess</code> (dev) / <code>nats</code> (prod)	<code>[IN-HOUSE: eventbus]</code>	NATS JetStream is the Large-scale transport.
<code>THREADY_NATS_URL</code>	url	unset	<code>[IN-HOUSE: eventbus]</code>	<code>nats://host:4222</code> .
<code>THREADY_NATS_STREAM</code>	string	<code>thready</code>	ASSUMPTION	JetStream stream name.
<code>THREADY_WORKERS</code>	int	<code>32</code>	<code>[DEFAULT – adjustable]</code> (Q4)	BackgroundTasks worker-pool size.
<code>THREADY_RETRY_MAX</code>	int	<code>5</code>	<code>[DEFAULT – adjustable]</code> (\$3.3)	Max retries per step/post.
<code>THREADY_RETRY_BASE</code>	duration	<code>2s</code>	<code>[DEFAULT – adjustable]</code>	Exponential back-off base.
<code>THREADY_RETRY_FACTOR</code>	float	<code>2.0</code>	<code>[DEFAULT – adjustable]</code>	Back-off multiplier.
<code>THREADY_RETRY_CAP</code>	duration	<code>5m</code>	<code>[DEFAULT – adjustable]</code>	Back-off ceiling.
<code>THREADY_POST_TIMEOUT</code>	duration	<code>30m</code>	<code>[DEFAULT – adjustable]</code>	Per-post soft budget (research-heavy).
<code>THREADY_SKILL_CONCURRENCY</code>	int	<code>8</code>	<code>[DEFAULT – adjustable]</code>	Per-Skill concurrency cap.

Idempotency (VERIFIED design). Each post is claimed exactly once via a Postgres row/advisory lock in the BackgroundTasks queue, so a “new post” event storm never double-processes a post `[CONSTITUTION §11.4.176]`. No env var disables this.

11. Authentication & security [GAP: 10]

VERIFIED: `digital.vasic.auth` defaults JWT to **HMAC-SHA256**, which is fine single-service but multi-service verification wants asymmetric keys. Thready plans **RS256/EdDSA + JWKS rotation** (`P1`). Set `THREADY_JWT_SIGNING_ALG=RS256` once the keypair is provisioned.

Variable	Type	Default	Provenance	Notes
THREADY_JWT_SIGNING_ALG	enum HS256\ RS256\ EdDSA	HS256 (dev) → RS256 (prod target)	[IN-HOUSE: auth] [GAP: 10]	
THREADY_JWT_SECRET	secret	unset	[IN-HOUSE: auth]	HS256 only; ≥32 bytes.
THREADY_JWT_PRIVATE_KEY_PATH / _PUBLIC_KEY_PATH	path	unset	[IN-HOUSE: auth]	RS256/EdDSA PEMs.
THREADY_ACCESS_TOKEN_TTL	duration	15m	[DEFAULT – adjustable] (Q9)	Access token lifetime.
THREADY_REFRESH_TOKEN_TTL	duration	168h (7 d)	[DEFAULT – adjustable]	Refresh token lifetime.
THREADY_IDLE_TIMEOUT	duration	30m	[DEFAULT – adjustable]	Web idle logout.
THREADY_MFA_REQUIRED_TIERS	csv	root,account_admin	[DEFAULT – adjustable] (Q9)	TOTP mandatory for admin tiers, optional for users.
THREADY_PASSWORD_MIN_LEN	int	12	[DEFAULT – adjustable]	Argon2id-hashed; breach-list checked.
THREADY_ARGON2_MEMORY_KIB	int	65536	[IN-HOUSE: security]	Argon2id KDF memory.
THREADY_API_KEY_HASH_PEPPER	secret	unset	[IN-HOUSE: auth]	Server-side pepper for API-key hashing.
THREADY_ENCRYPTION_KEY	secret (32 B)	unset	[IN-HOUSE: security]	AES-256-GCM master key for sealed storage. Redacted.

12. Downloads & 3rd-party systems [GAP: 4] [GAP: 5]

Variable	Type	Default	Provenance	Notes
THREADY_BOBA_URL	url	unset	[IN-HOUSE: Boba]	Torrent search/download (SSE + <code>POST /api/v1/hooks</code>).
THREADY_BOBA_CALLBACK_URL	url	derived	[GAP: 6.4]	Where Boba posts completion; contract being standardized.
THREADY_METUBE_URL	url	unset	[IN-HOUSE: MeTube]	Video/streaming download.
THREADY_METUBE_WEBHOOK_URL	url	derived	[BUILD-NEW] [GAP: 5]	MeTube is poll-only today ; outbound webhook is being added. Until then Thready polls <code>GET /api/postprocess/status</code> .
THREADY_DOWNLOAD_MANAGER_URL	url	unset	[BUILD-NEW] [GAP: 4]	Generic multi-protocol Download Manager does not exist yet ; HTTP(S) source + queue/resume semantics are being built.
THREADY_DOWNLOAD_CONCURRENCY	int	4	[DEFAULT – adjustable]	Parallel download jobs.
THREADY_GAME_DEFAULT_PLATFORMS	csv	PC-Windows,PS4,Android	[DEFAULT – adjustable]	<code>#Game</code> default targets (\$3.2.2).
THREADY_SOFTWARE_DEFAULT_OS	csv	Windows,Linux,macOS	[DEFAULT – adjustable]	<code>#Software</code> default OSes.

13. Assets & media directories

Directories are configurable and modifiable at runtime via the management API (client → REST → System).

Variable	Type	Default	Provenance	Notes
THREADY_ASSET_SERVICE_URL	url	unset	[BUILD-NEW] (Catalogizer) [GAP: 9]	Asset Service. Decoupling from Catalogizer and the HLS/DASH transcoder integration are P1 — Catalogizer's Streaming submodule is a WebSocket hub, not media/transcode streaming. Zero version serves raw + one -web rendition over HTTP Range. Client links resolve through this service, never direct paths.
THREADY_MEDIA_DIR	path	./data/media	[OPERATOR-editable]	Root for downloaded media.
THREADY_WEB_RENDITION_SUFFIX	string	-web	[DEFAULT – adjustable] (Q36)	Suffix before extension for web-optimized renditions.
THREADY_ENCRYPTED_ASSET_DIR	path	./data/secure	[IN-HOUSE: security]	AES-256-GCM directory for cards/contracts/QR/screenshots.
THREADY_ASSET_DEDUP	bool	true	[GAP: 6.1]	Content-hash dedup + integrity checksums.

14. Observability, logging & backup

`digital.vasic.observability` = OpenTelemetry + Prometheus + logrus + ClickHouse (VERIFIED; **not** ELK/Loki/Datadog).

Variable	Type	Default	Provenance	Notes
<code>OTEL_EXPORTER_OTLP_ENDPOINT</code>	url	unset	[IN-HOUSE: observability]	OTLP collector (Jaeger/Zipkin behind it).
<code>THREADY_METRICS_ADDR</code>	host:port	<code>0.0.0.0:9090</code>	[IN-HOUSE: observability]	Prometheus scrape endpoint.
<code>THREADY_CLICKHOUSE_DSN</code>	dsn	unset	[IN-HOUSE: observability]	Log/analytics sink.
<code>THREADY_AUDIT_RETENTION</code>	duration	<code>8760h</code> (1 y)	[DEFAULT — adjustable] (Q40)	Append-only audit-log retention.
<code>THREADY_BACKUP_FULL_CRON</code>	cron	<code>0 3 * * *</code>	[OPERATOR] (Q41)	Daily full backup.
<code>THREADY_BACKUP_INCREMENTAL_CRON</code>	cron	<code>0 * * * *</code>	[OPERATOR]	Hourly DB incrementals (RPO ≈ 1 h).
<code>FIREBASE_PROJECT_ID</code>	string	unset	[CONSTITUTION §11.4.47]	Crashlytics/ Analytics/ Distribution (mobile).

15. Retention, billing & localization

Variable	Type	Default	Provenance	Notes
THREADY_RETENTION_DEFAULT	duration indefinite	indefinite	[OPERATOR] (Q12)	Global default; Accounts may shorten.
THREADY_BILLING_MODE	enum subscription subscription+metered metered	subscription+metered	[OPERATOR] (Q11)	Metering on from day one.
THREADY_METERING_FLUSH	duration	1m	[DEFAULT – adjustable]	Usage-event flush interval.
THREADY_DEFAULT_LOCALE	enum en ru sr-Cyrl	en	[DEFAULT – adjustable] (Q35)	UI locale default.
THREADY_TRANSLATE_URL	url	unset	[IN-HOUSE: HelixTranslate]	On-demand translation service.

16. White-labeling & branding

Per-account branding; new accounts default to Thready/Helix Development branding (final request §8.3).

Variable	Type	Default	Provenance	Notes
<code>THREADY_BRAND_NAME</code>	string	<code>Thready</code>	<code>[OPERATOR-editable]</code>	System-default product name.
<code>THREADY_BRAND_PRIMARY_COLOR</code>	hex	<code>#B6E376</code>	<code>[IN-HOUSE: design_system]</code>	Helix-green base; per-account override in DB.
<code>THREADY_BRAND_LOGO_PATH</code>	path	<code>./assets/Logo.png</code>	<code>[OPERATOR-editable]</code>	Default logo (theme derived from it).
<code>THREADY_BRAND_SLOGAN</code>	string	<code>Made with love ♥ by Helix Development</code>	<code>[OPERATOR]</code>	Footer slogan (heart glyph); Helix attribution persists even under white-label.
<code>THREADY_THEME_DEFAULT</code>	enum <code>system</code> <code>light</code> <code>dark</code>	<code>system</code>	<code>[CONSTITUTION §11.4.162]</code>	Light + dark mandatory.

Note. Per-account branding overrides (colors/logo/slogan for a specific Account) are stored in the **database**, not env vars — they are edited by the Root Admin via the API/portal ([root-admin-guide.md §5](#)). Env vars set only the *system default* applied to new Accounts.

17. Precedence, validation & change management

- **Precedence:** process-env (shell export) > `.env` file > compiled default. Sourced secrets (`api_keys.sh`) and the private-repo mount populate process-env before the file is read.
- **Validation:** the service validates its required set at boot and **fails loudly** listing missing keys; use `thready config validate` ([cli-reference.md](#)) to check a `.env` without starting the service.
- **Runtime changes:** directory/frequency/branding settings changeable at runtime go **client** → **REST API** → **System** (final request §21.4) and are RBAC-gated; they persist to the DB and re-emit a `config.changed` event. Secrets are **not** hot-reloaded — rotating a key requires a service restart (documented in [troubleshooting.md](#)).
- **Docs sync:** `.env.example` is kept in lockstep with this reference via Docs Chain `[CONSTITUTION §11.4.65]`; a CI-equivalent local hook diffs the two.

18. Open items

- `[OPEN: cfg-1]` Final `THREADY_*` variable names are this document's proposal; ratify against the actual Thready service `config` package once it exists. Tracked workable item: **ATM — reconcile `THREADY_*` names with implemented config struct tags.**
- `[OPEN: cfg-2]` `THREADY_VECTOR_BACKEND=qdrant|pinecone|milvus` are unverified `[GAP: 8]`; keep on `pgvector` until the Qdrant backend is integration-tested. Workable item: **ATM — harden Qdrant backend to pgvector parity.**
- `[OPEN: cfg-3]` `THREADY_METUBE_WEBHOOK_URL` / `THREADY_DOWNLOAD_MANAGER_URL` reference `[BUILD-NEW]` services; their exact request/response contract is finalized in the callback-module spec. Workable item: **ATM — publish standardized callback schema.**
- `[OPEN: cfg-4]` Searchable-but-sealed credential representation not yet implemented `[GAP: 7]`; credential content is encrypted but not semantically indexed. Workable item: **ATM — redacted-token embedding over `securestorage`.**

Closed in Rev 2 (Pass 3):

- `[CLOSED: cfg-messenger]` Messenger env-var names were an ASSUMPTION. **Verified at source** in `vasic-digital/herald/.env.example` (+ `docs/guides/messengers/`): the real names are `HERALD_MTPROTO_APP_ID/APP_HASH/PHONE/PASSWORD/SESSION_FILE` (user client), `HERALD_TGRAM_BOT_TOKEN/CHAT_ID/LIVE_INBOUND` (Bot API), `HERALD_OPERATOR_IDS`, and the reserved `HERALD_MAX_BOT_TOKEN/CHAT_ID`. \$9 now carries the VERIFIED names.
- `[CLOSED: cfg-vision]` VisionEngine variables verified in `helix_track/vision_engine/.env.example` (`HELIX_VISION_OPENCV_ENABLED`, `HELIX_VISION_SSIM_THRESHOLD`, `HELIX_OLLAMA_*`, `HELIX_LLAMACPP_RPC_*`, `ASTICA_API_KEY`, provider enum widened) — \$8.3.
- `[CLOSED: cfg-containers]` Additional `CONTAINERS_REMOTE_*` verified in `helix_track/containers/.env.example` — \$5.

Appendix A — Master environment-variable index

This is the single **one-row-per-variable** consolidated index the original request mandates (“*All environment variables ... MUST be properly documented*”). Columns are **Name • Purpose • Default • Scope • Example**. Scope names which component reads the variable and in which environment it applies (`all` = every service/env unless noted). Provenance/verification detail for each row lives in the categorized section cross-referenced by the section number in the Name cell's group heading above; `(V)` marks a VERIFIED module variable, `(A)` an ASSUMPTION/`THREADY_*` proposal, `(D)` a `[DEFAULT — adjustable]`.

A.1 Core runtime, environments, TLS

Name	Purpose	Default	Scope	Example
<code>THREADY_ENV</code> (A)	Select env profile & safety rails	<code>development</code>	all services	<code>THREADY_ENV=production</code>
<code>THREADY_ENV_FILE</code> (A)	Explicit <code>.env</code> path	<code>./.env</code>	all services	<code>THREADY_ENV_FILE=/etc/thready/prod.env</code>
<code>THREADY_HTTP_ADDR</code> (A)	REST + WS/SSE bind	<code>0.0.0.0:8443</code>	API server	<code>THREADY_HTTP_ADDR=0.0.0.0:8443</code>
<code>THREADY_HTTP3_ENABLED</code> (V)	Enable HTTP/3 (QUIC)	<code>true</code>	API server	<code>THREADY_HTTP3_ENABLED=true</code>
<code>THREADY_HTTP_COMPRESSION</code> (D)	Response compression	<code>br,gzip</code>	API server	<code>THREADY_HTTP_COMPRESSION=br,gzip</code>
<code>THREADY_LOG_LEVEL</code> (V)	logrus level	<code>info</code>	all services	<code>THREADY_LOG_LEVEL=debug</code>
<code>THREADY_LOG_FORMAT</code> (V)	Log format	<code>json</code> (<code>text</code> dev)	all services	<code>THREADY_LOG_FORMAT=json</code>
<code>THREADY_REQUEST_TIMEOUT</code> (D)	Per-request server timeout	<code>30s</code>	API server	<code>THREADY_REQUEST_TIMEOUT=30s</code>
<code>THREADY_RATE_LIMIT_RPS</code> (V)	Per-identity rate cap	<code>100</code>	API server	<code>THREADY_RATE_LIMIT_RPS=100</code>
<code>THREADY_CORS_ORIGINS</code> (V)	Allowed browser origins	<code>(none)</code>	API server	<code>THREADY_CORS_ORIGINS=https://thready.hxd3v.com</code>
<code>THREADY_PORT_PREFIX</code> (V)	Deterministic dynamic-port base	<code>unset</code>	container stack	<code>THREADY_PORT_PREFIX=24</code>
<code>THREADY_PUBLIC_DOMAIN</code> (op)	Base domain	<code>thready.hxd3v.com</code>	all services	<code>THREADY_PUBLIC_DOMAIN=thready.hxd3v.com</code>
<code>THREADY_PUBLIC_BASE_URL</code> (A)	Absolute link base	<code>https://thready.hxd3v.com</code>	API/content	<code>THREADY_PUBLIC_BASE_URL=https://thready.hxd3v.com</code>
<code>LETS_ENCRYPT_EMAIL</code> (V)	ACME account email	<code>unset</code>	proxy/prod	<code>LETS_ENCRYPT_EMAIL=ops@hxd3v.com</code>
<code>LETS_ENCRYPT_CHALLENGE</code> (V)	ACME challenge type	<code>http-01</code>	proxy/prod	<code>LETS_ENCRYPT_CHALLENGE=dns-01</code>

Name	Purpose	Default	Scope	Example
<code>THREADY_TLS_MIN_VERSION</code> (D)	Minimum TLS	<code>1.3</code>	API/proxy	<code>THREADY_TLS_MIN_VERSION=1.3</code>
<code>CONTAINERS_REMOTE_ENABLE</code> (V)	Remote container distribution	<code>false</code>	deploy	<code>CONTAINERS_REMOTE_ENABLED=true</code>
<code>CONTAINERS_REMOTE_DEFAULT_SSH_USER</code> (V)	Deploy SSH user	<code>deploy</code>	deploy	<code>CONTAINERS_REMOTE_DEFAULT_SSH_USER=thready</code>
<code>CONTAINERS_REMOTE_DEFAULT_RUNTIME</code> (V)	Runtime	<code>podman</code>	deploy	<code>CONTAINERS_REMOTE_DEFAULT_RUNTIME=podman</code>
<code>CONTAINERS_REMOTE_DEFAULT_SSH_KEY</code> (V)	Deploy SSH key	<code>unset</code>	deploy	<code>CONTAINERS_REMOTE_DEFAULT_SSH_KEY=~/.ssh/thready_deploy</code>
<code>CONTAINERS_REMOTE_PORT_RANGE_START / _END</code> (V)	Tunnel/port range	<code>20000 / 30000</code>	deploy	<code>CONTAINERS_REMOTE_PORT_RANGE_START=20000</code>
<code>CONTAINERS_REMOTE_CONNECT_TIMEOUT</code> (V)	SSH connect timeout	<code>10s</code>	deploy	<code>CONTAINERS_REMOTE_CONNECT_TIMEOUT=10s</code>
<code>CONTAINERS_REMOTE_COMMAND_TIMEOUT</code> (V)	Remote command timeout	<code>5m</code>	deploy	<code>CONTAINERS_REMOTE_COMMAND_TIMEOUT=5m</code>
<code>CONTAINERS_REMOTE_SSH_CONTROL_MASTER</code> (V)	Reuse SSH master	<code>true</code>	deploy	<code>CONTAINERS_REMOTE_SSH_CONTROL_MASTER=true</code>
<code>CONTAINERS_REMOTE_SSH_CONTROL_PERSIST</code> (V)	SSH master linger	<code>60s</code>	deploy	<code>CONTAINERS_REMOTE_SSH_CONTROL_PERSIST=60s</code>
<code>CONTAINERS_REMOTE_SSH_MAX_CONNECTIONS</code> (V)	SSH concurrency/target	<code>10</code>	deploy	<code>CONTAINERS_REMOTE_SSH_MAX_CONNECTIONS=10</code>
	Multi-host placement	<code>roundrobin</code>	deploy	<code>CONTAINERS_REMOTE_SCHEDULER=leastloaded</code>

Name	Purpose	Default	Scope	Example
<code>CONTAINERS_REMOTE_SCHEDULER</code> (V)				
<code>CONTAINERS_REMOTE_VOLUME_TYPE</code> (V)	Default mount type	<code>volume</code>	deploy	<code>CONTAINERS_REMOTE_VOLUME_TYPE=volume</code>

A.2 Data, datastores, embeddings

Name	Purpose	Default	Scope	Example
<code>THREADY_DB_DRIVER</code> (V)	Relational backend	<code>sqlite</code> / <code>postgres</code>	all services	<code>THREADY_DB_DRIVER=postgres</code>
<code>THREADY_DB_DSN</code> (V)	DB connection string	(see quick-start)	all services	<code>THREADY_DB_DSN=postgres://thready:***@db:5432/thready</code>
<code>THREADY_DB_MAX_OPEN_CONNS</code> (D)	pgx pool size	<code>32</code>	all services	<code>THREADY_DB_MAX_OPEN_CONNS=64</code>
<code>THREADY_DB_MAX_IDLE_CONNS</code> (D)	Idle pool	<code>8</code>	all services	<code>THREADY_DB_MAX_IDLE_CONNS=8</code>
<code>THREADY_DB_CONN_MAX_LIFETIME</code> (D)	Connection recycle	<code>30m</code>	all services	<code>THREADY_DB_CONN_MAX_LIFETIME=30m</code>
<code>THREADY_DB_MIGRATE_ON_BOOT</code> (V)	Auto-migrate	<code>true</code> dev/ <code>false</code> prod	API/migrator	<code>THREADY_DB_MIGRATE_ON_BOOT=false</code>
<code>THREADY_DB_PARTITIONING</code> (gap)	Time-partition <code>posts</code>	<code>true</code> prod	DB	<code>THREADY_DB_PARTITIONING=true</code>
<code>THREADY_VECTOR_BACKEND</code> (V)	Vector store	<code>pgvector</code>	search/workers	<code>THREADY_VECTOR_BACKEND=pgvector</code>
<code>THREADY_VECTOR_DSN</code> (V)	Vector DB DSN	(shares Postgres)	search/workers	<code>THREADY_VECTOR_DSN=postgres://thready:***@db:5432/thready</code>
<code>THREADY_VECTOR_METRIC</code> (V)	Distance metric	<code>cosine</code>	search	<code>THREADY_VECTOR_METRIC=cosine</code>
<code>THREADY_EMBEDDING_DIM</code> (gap)	Embedding dimension	<code>1024</code>	search/workers	<code>THREADY_EMBEDDING_DIM=1024</code>
<code>THREADY_VECTOR_INDEX</code> (D)	ANN index	<code>hnsw</code>	search	<code>THREADY_VECTOR_INDEX=hnsw</code>
<code>THREADY_QDRANT_URL</code> (gap)	Qdrant endpoint (unverified)	unset	search	<code>THREADY_QDRANT_URL=http://qdrant:6333</code>
	Cache tier	<code>memory</code> / <code>redis</code>	all services	

Name	Purpose	Default	Scope	Example
<code>THREADY_CACHE_BACKEND (V)</code>				<code>THREADY_CACHE_BACKEND=redis</code>
<code>THREADY_CACHE_REDIS_URL (V)</code>	Redis URL	unset	all services	<code>THREADY_CACHE_REDIS_URL=redis://cache:6379/0</code>
<code>THREADY_CACHE_TTL (D)</code>	Entry TTL	<code>10m</code>	all services	<code>THREADY_CACHE_TTL=10m</code>
<code>THREADY_STORAGE_BACKEND (V)</code>	Asset store	<code>filesystem / minio</code>	asset service	<code>THREADY_STORAGE_BACKEND=minio</code>
<code>THREADY_STORAGE_ENDPOINT (V)</code>	MinIO endpoint	unset	asset service	<code>THREADY_STORAGE_ENDPOINT=https://minio:9000</code>
<code>THREADY_STORAGE_BUCKET (A)</code>	Object bucket	<code>thready-assets</code>	asset service	<code>THREADY_STORAGE_BUCKET=thready-assets</code>
<code>THREADY_STORAGE_ACCESS_KEY (V)</code>	Storage key (secret)	unset	asset service	<code>THREADY_STORAGE_ACCESS_KEY=***</code>
<code>THREADY_STORAGE_SECRET_KEY (V)</code>	Storage secret	unset	asset service	<code>THREADY_STORAGE_SECRET_KEY=***</code>
<code>THREADY_STORAGE_SIGNED_URL_TTL (gap)</code>	Signed-URL lifetime	<code>15m</code>	asset service	<code>THREADY_STORAGE_SIGNED_URL_TTL=15m</code>
<code>HELIX_EMBEDDING_PROVIDER (V)</code>	Embedder — must be real	<code>llama</code> (enforced)	search/workers	<code>HELIX_EMBEDDING_PROVIDER=llama</code>
<code>THREADY_EMBEDDING_BASE_URL (V)</code>	Embedding endpoint	<code>http://localhost:8080/v1</code>	search/workers	<code>THREADY_EMBEDDING_BASE_URL=http://llm:8080/v1</code>
<code>THREADY_EMBEDDING_MODEL (R)</code>	Embedding model	<code>jina-embeddings-v2-base-code</code>	search/workers	<code>THREADY_EMBEDDING_MODEL=jina-embeddings-v2-base-code</code>
<code>THREADY_EMBEDDING_API_KEY (V)</code>	Embedding key	unset	search/workers	<code>THREADY_EMBEDDING_API_KEY=***</code>

A.3 LLM, vision, messengers, downloads

Name	Purpose	Default	Scope	Example
<code>HELIX_LLM_BASE_URL</code> (V)	Local llama.cpp URL	unset	workers	<code>HELIX_LLM_BASE_URL=http://llm:8080</code>
<code>HELIX_LLM_MODEL</code> (V)	Reasoning model	unset	workers	<code>HELIX_LLM_MODEL=Llama-3.1-70B-Instruct-Q4_K_M</code>
<code>HELIX_LLM_CODE_MODEL</code> (V)	Code model	unset	workers	<code>HELIX_LLM_CODE_MODEL=Qwen2.5-Coder</code>
<code>ANTHROPIC_API_KEY</code> / <code>OPENAI_API_KEY</code> / <code>GOOGLE_API_KEY</code> / <code>DEEPSEEK_API_KEY</code> (V)	Cloud LLM fallbacks	unset	workers	<code>OPENAI_API_KEY=sk-***</code>
<code>GROQ_</code> / <code>MISTRAL_</code> / <code>QWEN_</code> / <code>OPENROUTER_</code> / <code>COHERE_</code> / <code>TOGETHER_</code> / <code>XAI_</code> / <code>PERPLEXITY_</code> / <code>NVIDIA_API_KEY</code> (V)	More LLM adapters	unset	workers	<code>GROQ_API_KEY=***</code>
<code>CEREBRAS_</code> / <code>FIREWORKS_</code> / <code>HUGGINGFACE_</code> / <code>SAMBANOVA_</code> / <code>SILICONFLOW_API_KEY</code> (V)	More LLM adapters	unset	workers	<code>FIREWORKS_API_KEY=***</code>
<code>REPLICATE_API_TOKEN</code> (V)	Replicate adapter (note <code>_TOKEN</code>)	unset	workers	<code>REPLICATE_API_TOKEN=r8_***</code>
<code>THREADY_LLM_MAX_RETRIES</code> (D)	Provider retry ceiling	5	workers	<code>THREADY_LLM_MAX_RETRIES=5</code>
<code>THREADY_LLM_CIRCUIT_BREAKER</code> (V)	Break flapping providers	true	workers	<code>THREADY_LLM_CIRCUIT_BREAKER=true</code>
<code>HELIX_VISION_PROVIDER</code> (V)	Vision provider select	auto	vision workers	<code>HELIX_VISION_PROVIDER=auto</code>

Name	Purpose	Default	Scope	Example
<code>HELIX_VISION_TIMEOUT</code> (V)	Vision timeout (s)	<code>60</code>	vision workers	<code>HELIX_VISION_TIMEOUT=60</code>
<code>HELIX_VISION_MAX_IMAGE_SIZE</code> (V)	Downscale ceiling (px)	<code>4096</code>	vision workers	<code>HELIX_VISION_MAX_IMAGE_SIZE=4096</code>
<code>HELIX_VISION_OPENCV_ENABLED</code> (V)	OpenCV pre-pass	<code>true</code>	vision workers	<code>HELIX_VISION_OPENCV_ENABLED=true</code>
<code>HELIX_VISION_SSIM_THRESHOLD</code> (V)	Duplicate cutoff	<code>0.95</code>	vision workers	<code>HELIX_VISION_SSIM_THRESHOLD=0.95</code>
<code>HELIX_VISION_HOSTS</code> / <code>HELIX_VISION_USER</code> (V)	Remote vision fleet	<code>unset</code>	vision workers	<code>HELIX_VISION_HOSTS=tinker.local,amber.local</code>
<code>HELIX_OLLAMA_URL</code> / <code>HELIX_OLLAMA_MODEL</code> (V)	Local Ollama vision	<code>...:11434</code> / <code>minicpm-v:8b</code>	vision workers	<code>HELIX_OLLAMA_MODEL=minicpm-v:8b</code>
<code>HELIX_LLAMACPP_RPC_ENABLED</code> / <code>_WORKERS</code> / <code>_MODEL</code> (V)	llama.cpp RPC vision	<code>false</code> / <code>unset</code> / <code>~/models/...</code>	vision workers	<code>HELIX_LLAMACPP_RPC_ENABLED=false</code>
<code>ASTICA_API_KEY</code> / <code>KIMI_API_KEY</code> / <code>STEPFUN_API_KEY</code> (V)	Vision provider keys	<code>unset</code>	vision workers	<code>ASTICA_API_KEY=***</code>
<code>THREADY_OCR_PROVIDER</code> (BN)	OCR engine (pending)	<code>none</code>	vision workers	<code>THREADY_OCR_PROVIDER=none</code>
<code>THREADY_OCR_LANGS</code> (D)	OCR language packs	<code>eng,rus</code>	vision workers	<code>THREADY_OCR_LANGS=eng,rus,srp</code>
<code>HERALD_MTPROTO_APP_ID</code> (V)	Telegram user app_id	<code>unset</code>	ingest	<code>HERALD_MTPROTO_APP_ID=1234567</code>
<code>HERALD_MTPROTO_APP_HASH</code> (V)	Telegram user app_hash	<code>unset</code>	ingest	<code>HERALD_MTPROTO_APP_HASH=abcd...32hex</code>
	Reading account phone	<code>unset</code>	ingest	

Name	Purpose	Default	Scope	Example
HERALD_MTPROTO_PHONE (V)				HERALD_MTPROTO_PHONE =+15551234567
HERALD_MTPROTO_PASSWORD (V)	Cloud 2FA password	unset	ingest	HERALD_MTPROTO_PASSWORD=***
HERALD_MTPROTO_SESSION_FILE (V)	Session path	~/.config/herald/ mtpROTO.session	ingest	HERALD_MTPROTO_SESSION_FILE=/secure/ mtpROTO.session
HERALD_TGRAM_BOT_TOKEN (V)	Bot-API reply token	unset	reply	HERALD_TGRAM_BOT_TOKEN=123:AA***
HERALD_TGRAM_CHAT_ID (V)	Bot-API target chat	unset	reply	HERALD_TGRAM_CHAT_ID=-1001234567890
HERALD_TGRAM_LIVE_INBOUND (V)	Bot inbound long-poll	false	reply	HERALD_TGRAM_LIVE_INBOUND=true
HERALD_OPERATOR_IDS (V)	Operator user allowlist	unset	reply	HERALD_OPERATOR_IDS=111,222
HERALD_MAX_BOT_TOKEN / HERALD_MAX_CHAT_ID (V, reserved)	Max (PLANNED)	unset	(none yet)	HERALD_MAX_BOT_TOKEN= =
THREADY_MESSENGER_SIGNIN_MODE (op)	Interactive vs headless	interactive	ingest	THREADY_MESSENGER_SIGNIN_MODE=noninteractive
THREADY_POLL_INTERVAL (D)	Poll frequency	5m	ingest	THREADY_POLL_INTERVAL=2m
THREADY_REPLY_ACCOUNT (D)	Who posts replies	robot	reply	THREADY_REPLY_ACCOUNT=T=robot
THREADY_BOBA_URL (V)	Torrent service	unset	workers	THREADY_BOBA_URL=http://boba:8000
THREADY_BOBA_CALLBACK_URL (gap)	Boba completion callback	derived	workers	THREADY_BOBA_CALLBACK_URL=https:// thready.hxd3v.com/ v1/hooks/boba
THREADY_METUBE_URL (V)	Video downloader	unset	workers	THREADY_METUBE_URL=http://metube:8081

Name	Purpose	Default	Scope	Example
<code>THREASY_METUBE_WEBHOOK_URL</code> (BN)	MeTube webhook (pending)	derived	workers	<code>THREASY_METUBE_WEBHOOK_URL=https://thready.hxd3v.com/v1/hooks/metube</code>
<code>THREASY_DOWNLOAD_MANAGER_URL</code> (BN)	Generic DL manager (pending)	unset	workers	<code>THREASY_DOWNLOAD_MANAGER_URL=http://d1m:8082</code>
<code>THREASY_DOWNLOAD_CONCURRENCY</code> (D)	Parallel downloads	<code>4</code>	workers	<code>THREASY_DOWNLOAD_CONCURRENCY=4</code>
<code>THREASY_GAME_DEFAULT_PLATFORMS</code> (D)	<code>#Game</code> targets	<code>PC-Windows,PS4,Android</code>	workers	<code>THREASY_GAME_DEFAULT_PLATFORMS=PC-Windows,PS4</code>
<code>THREASY_SOFTWARE_DEFAULT_OS</code> (D)	<code>#Software</code> OSes	<code>Windows,Linux,macOS</code>	workers	<code>THREASY_SOFTWARE_DEFAULT_OS=Windows,Linux</code>

A.4 Event bus, processing, auth, assets, observability, retention, branding

Name	Purpose	Default	Scope	Example
<code>THREADY_EVENTBUS_BACKEND</code> (V)	Event transport	<code>inprocess</code> / <code>nats</code>	all services	<code>THREADY_EVENTBUS_BACKEND=nats</code>
<code>THREADY_NATS_URL</code> (V)	NATS URL	unset	all services	<code>THREADY_NATS_URL=nats://nats:4222</code>
<code>THREADY_NATS_STREAM</code> (A)	JetStream stream	<code>thready</code>	all services	<code>THREADY_NATS_STREAM=thready</code>
<code>THREADY_WORKERS</code> (D)	Worker-pool size	<code>32</code>	workers	<code>THREADY_WORKERS=64</code>
<code>THREADY_RETRY_MAX</code> (D)	Max retries/post	<code>5</code>	workers	<code>THREADY_RETRY_MAX=5</code>
<code>THREADY_RETRY_BASE</code> (D)	Back-off base	<code>2s</code>	workers	<code>THREADY_RETRY_BASE=2s</code>
<code>THREADY_RETRY_FACTOR</code> (D)	Back-off multiplier	<code>2.0</code>	workers	<code>THREADY_RETRY_FACTOR=2.0</code>
<code>THREADY_RETRY_CAP</code> (D)	Back-off ceiling	<code>5m</code>	workers	<code>THREADY_RETRY_CAP=5m</code>
<code>THREADY_POST_TIMEOUT</code> (D)	Per-post soft budget	<code>30m</code>	workers	<code>THREADY_POST_TIMEOUT=30m</code>
<code>THREADY_SKILL_CONCURRENCY</code> (D)	Per-Skill concurrency	<code>8</code>	workers	<code>THREADY_SKILL_CONCURRENCY=8</code>
<code>THREADY_JWT_SIGNING_ALG</code> (V)	JWT algorithm	<code>HS256</code> → <code>RS256</code>	auth/all	<code>THREADY_JWT_SIGNING_ALG=RS256</code>
<code>THREADY_JWT_SECRET</code> (V)	HS256 secret	unset	auth/all	<code>THREADY_JWT_SECRET=***≥32B</code>
<code>THREADY_JWT_PRIVATE_KEY_PATH</code> / <code>THREADY_JWT_PUBLIC_KEY_PATH</code> (V)	RS256/EdDSA PEMs	unset	auth/all	<code>THREADY_JWT_PRIVATE_KEY_PATH=/secure/jwt.pem</code>
<code>THREADY_ACCESS_TOKEN_TTL</code> (D)	Access token life	<code>15m</code>	auth	<code>THREADY_ACCESS_TOKEN_TTL=15m</code>
<code>THREADY_REFRESH_TOKEN_TTL</code> (D)	Refresh token life	<code>168h</code>	auth	<code>THREADY_REFRESH_TOKEN_TTL=168h</code>

Name	Purpose	Default	Scope	Example
<code>THREADY_IDLE_TIMEOUT</code> (D)	Web idle logout	<code>30m</code>	API/web	<code>THREADY_IDLE_TIMEOUT=30m</code>
<code>THREADY_MFA_REQUIRED_TIERS</code> (D)	Tiers forced to MFA	<code>root,account_admin</code>	auth	<code>THREADY_MFA_REQUIRED_TIERS=root,account_admin</code>
<code>THREADY_PASSWORD_MIN_LEN</code> (D)	Min password length	<code>12</code>	auth	<code>THREADY_PASSWORD_MIN_LEN=14</code>
<code>THREADY_ARGON2_MEMORY_KIB</code> (V)	Argon2id memory	<code>65536</code>	auth	<code>THREADY_ARGON2_MEMORY_KIB=65536</code>
<code>THREADY_API_KEY_HASH_PEPPER</code> (V)	API-key pepper	<code>unset</code>	auth	<code>THREADY_API_KEY_HASH_PEPPER=***</code>
<code>THREADY_ENCRYPTION_KEY</code> (V)	AES-256-GCM master	<code>unset</code>	security/all	<code>THREADY_ENCRYPTION_KEY=***32B</code>
<code>THREADY_ASSET_SERVICE_URL</code> (BN)	Asset Service (pending decouple)	<code>unset</code>	clients/workers	<code>THREADY_ASSET_SERVICE_URL=http://assets:8090</code>
<code>THREADY_MEDIA_DIR</code> (op)	Media root	<code>./data/media</code>	asset service	<code>THREADY_MEDIA_DIR=./data/media</code>
<code>THREADY_WEB_RENDITION_SUFFIX</code> (D)	<code>-web</code> rendition suffix	<code>-web</code>	asset service	<code>THREADY_WEB_RENDITION_SUFFIX=-web</code>
<code>THREADY_ENCRYPTED_ASSET_DIR</code> (V)	Sealed asset dir	<code>./data/secure</code>	asset service	<code>THREADY_ENCRYPTED_ASSET_DIR=./data/secure</code>
<code>THREADY_ASSET_DEDUP</code> (gap)	Content-hash dedup	<code>true</code>	asset service	<code>THREADY_ASSET_DEDUP=true</code>
<code>OTEL_EXPORTER_OTLP_ENDPOINT</code> (V)	OTLP collector	<code>unset</code>	all services	<code>OTEL_EXPORTER_OTLP_ENDPOINT=http://otel:4317</code>
<code>THREADY_METRICS_ADDR</code> (V)	Prometheus endpoint	<code>0.0.0.0:9090</code>	all services	<code>THREADY_METRICS_ADDR=0.0.0.0:9090</code>
<code>THREADY_CLICKHOUSE_DSN</code> (V)	Log/analytics sink	<code>unset</code>	all services	<code>THREADY_CLICKHOUSE_DSN=clickhouse://ch:9000/thready</code>

Name	Purpose	Default	Scope	Example
<code>THREADY_AUDIT_RETENTION</code> (D)	Audit retention	<code>8760h</code>	audit	<code>THREADY_AUDIT_RETENTION=8760h</code>
<code>THREADY_BACKUP_FULL_CRON</code> (op)	Full backup schedule	<code>0 3 * * *</code>	backup	<code>THREADY_BACKUP_FULL_CRON=0 3 * * *</code>
<code>THREADY_BACKUP_INCREMENTAL_CRON</code> (op)	Incremental schedule	<code>0 * * * *</code>	backup	<code>THREADY_BACKUP_INCREMENTAL_CRON=0 * * * *</code>
<code>FIREBASE_PROJECT_ID</code> (V)	Crashlytics/Distribution	unset	mobile	<code>FIREBASE_PROJECT_ID=thready-prod</code>
<code>THREADY_RETENTION_DEFAULT</code> (op)	Global retention	<code>indefinite</code>	all services	<code>THREADY_RETENTION_DEFAULT=indefinite</code>
<code>THREADY_BILLING_MODE</code> (op)	Billing model	<code>subscription+metered</code>	billing	<code>THREADY_BILLING_MODE=subscription+metered</code>
<code>THREADY_METERING_FLUSH</code> (D)	Meter flush interval	<code>1m</code>	billing	<code>THREADY_METERING_FLUSH=1m</code>
<code>THREADY_DEFAULT_LOCALE</code> (D)	UI locale default	<code>en</code>	web/mobile	<code>THREADY_DEFAULT_LOCALE=sr-Cyrl</code>
<code>THREADY_TRANSLATE_URL</code> (V)	HelixTranslate URL	unset	web/workers	<code>THREADY_TRANSLATE_URL=http://translate:8095</code>
<code>THREADY_BRAND_NAME</code> (op)	System brand name	<code>Thready</code>	web/content	<code>THREADY_BRAND_NAME=Thready</code>
<code>THREADY_BRAND_PRIMARY_COLOR</code> (V)	Brand base color	<code>#B6E376</code>	web/content	<code>THREADY_BRAND_PRIMARY_COLOR=#B6E376</code>
<code>THREADY_BRAND_LOGO_PATH</code> (op)	Default logo	<code>./assets/Logo.png</code>	web/content	<code>THREADY_BRAND_LOGO_PATH=./assets/Logo.png</code>
<code>THREADY_BRAND_SLOGAN</code> (op)	Footer slogan	<code>Made with love ❤️ by Helix Development</code>	web/content	<i>(as default)</i>
<code>THREADY_THEME_DEFAULT</code> (V)	Default theme	<code>system</code>	web/mobile	<code>THREADY_THEME_DEFAULT=system</code>

(*op*) = `[OPERATOR]`, (*R*) = `[RESEARCH]`, (*gap*) = tied to a gap-register item, (*BN*) = `[BUILD-NEW]` (service not shipped; variable reserved for when it lands).

Appendix B — Worked `.env` examples per environment

Three complete, copy-pasteable skeletons — **development**, **staging**, **production** — showing the same variables at the settings each environment actually uses. Secrets are placeholders; never commit real values (`chmod 600 .env`).

B.1 Development (SQLite, in-process bus, local llama.cpp)

```

THREADY_ENV=development
THREADY_HTTP_ADDR=0.0.0.0:8443
THREADY_LOG_LEVEL=debug
THREADY_LOG_FORMAT=text
THREADY_DB_DRIVER=sqlite
THREADY_DB_DSN=file:./data/thready.db?_pragma=busy_timeout(5000)
THREADY_DB_MIGRATE_ON_BOOT=true
THREADY_VECTOR_BACKEND=pgvector
THREADY_VECTOR_DSN=postgres://thready:thready@localhost:5432/thready?sslmode=disable
THREADY_EMBEDDING_DIM=1024
HELIX_EMBEDDING_PROVIDER=llama
THREADY_EMBEDDING_BASE_URL=http://localhost:8080/v1
THREADY_EMBEDDING_MODEL=jina-embeddings-v2-base-code
THREADY_EVENTBUS_BACKEND=inprocess
THREADY_CACHE_BACKEND=memory
THREADY_STORAGE_BACKEND=filesystem
THREADY_MEDIA_DIR=./data/media
THREADY_JWT_SIGNING_ALG=HS256
THREADY_JWT_SECRET=dev-only-change-me-32-bytes-minimum
HERALD_MTPROTO_APP_ID=
HERALD_MTPROTO_APP_HASH=
HERALD_MTPROTO_PHONE=

```

B.2 Staging (Postgres, NATS, MinIO — mirrors prod, smaller)

```
THREADY_ENV=staging
THREADY_HTTP_ADDR=0.0.0.0:8443
THREADY_LOG_LEVEL=info
THREADY_LOG_FORMAT=json
THREADY_PUBLIC_DOMAIN=sta.thready.hxd3v.com
THREADY_DB_DRIVER=postgres
THREADY_DB_DSN=postgres://thready:${DB_PW}@sta-db:5432/thready?sslmode=require
THREADY_DB_MIGRATE_ON_BOOT=false
THREADY_VECTOR_BACKEND=pgvector
THREADY_VECTOR_DSN=postgres://thready:${DB_PW}@sta-db:5432/thready?sslmode=require
THREADY_EMBEDDING_DIM=1024
HELIX_EMBEDDING_PROVIDER=llama
THREADY_EMBEDDING_BASE_URL=http://llm:8080/v1
THREADY_EVENTBUS_BACKEND=nats
THREADY_NATS_URL=nats://sta-nats:4222
THREADY_CACHE_BACKEND=redis
THREADY_CACHE_REDIS_URL=redis://sta-cache:6379/0
THREADY_STORAGE_BACKEND=minio
THREADY_STORAGE_ENDPOINT=https://sta-minio:9000
THREADY_STORAGE_BUCKET=thready-assets-sta
THREADY_JWT_SIGNING_ALG=RS256
THREADY_JWT_PRIVATE_KEY_PATH=/secure/jwt-sta.pem
THREADY_JWT_PUBLIC_KEY_PATH=/secure/jwt-sta.pub.pem
THREADY_MFA_REQUIRED_TIERS=root,account_admin
OTEL_EXPORTER_OTLP_ENDPOINT=http://sta-otel:4317
THREADY_CLICKHOUSE_DSN=clickhouse://sta-ch:9000/thready
HERALD_MTPROTO_SESSION_FILE=/secure/mtproto-sta.session
THREADY_MESSENGER_SIGNIN_MODE=noninteractive
```

B.3 Production (hardened; RS256, backups, partitioning)

```

THREADY_ENV=production
THREADY_HTTP_ADDR=0.0.0.0:8443
THREADY_HTTP3_ENABLED=true
THREADY_LOG_LEVEL=info
THREADY_LOG_FORMAT=json
THREADY_PUBLIC_DOMAIN=thready.hxd3v.com
THREADY_TLS_MIN_VERSION=1.3
LETS_ENCRYPT_EMAIL=ops@hxd3v.com
THREADY_DB_DRIVER=postgres
THREADY_DB_DSN=postgres://thready:${DB_PW}@db:5432/thready?sslmode=require
THREADY_DB_MIGRATE_ON_BOOT=false
THREADY_DB_MAX_OPEN_CONNS=64
THREADY_DB_PARTITIONING=true
THREADY_VECTOR_BACKEND=pgvector
THREADY_VECTOR_DSN=postgres://thready:${DB_PW}@db:5432/thready?sslmode=require
THREADY_EMBEDDING_DIM=1024
THREADY_VECTOR_INDEX=hnsf
HELIX_EMBEDDING_PROVIDER=llama
THREADY_EMBEDDING_BASE_URL=http://llm:8080/v1
THREADY_EVENTBUS_BACKEND=nats
THREADY_NATS_URL=nats://nats:4222
THREADY_WORKERS=64
THREADY_CACHE_BACKEND=redis
THREADY_CACHE_REDIS_URL=redis://cache:6379/0
THREADY_STORAGE_BACKEND=minio
THREADY_STORAGE_ENDPOINT=https://minio:9000
THREADY_STORAGE_BUCKET=thready-assets
THREADY_JWT_SIGNING_ALG=RS256
THREADY_JWT_PRIVATE_KEY_PATH=/secure/jwt.pem
THREADY_JWT_PUBLIC_KEY_PATH=/secure/jwt.pub.pem
THREADY_ACCESS_TOKEN_TTL=15m
THREADY_REFRESH_TOKEN_TTL=168h
THREADY_MFA_REQUIRED_TIERS=root,account_admin
THREADY_ENCRYPTION_KEY=${AES_MASTER_KEY}
THREADY_ENCRYPTED_ASSET_DIR=/data/secure
OTEL_EXPORTER_OTLP_ENDPOINT=http://otel:4317
THREADY_METRICS_ADDR=0.0.0.0:9090
THREADY_CLICKHOUSE_DSN=clickhouse://ch:9000/thready
THREADY_AUDIT_RETENTION=8760h
THREADY_BACKUP_FULL_CRON=0 3 * * *
THREADY_BACKUP_INCREMENTAL_CRON=0 * * * *
THREADY_RETENTION_DEFAULT=indefinite
THREADY_BILLING_MODE=subscription+metered
THREADY_DEFAULT_LOCALE=en
THREADY_MESSENGER_SIGNIN_MODE=noninteractive
HERALD_MTPROTO_SESSION_FILE=/secure/mtproto.session

```

Made with love ♥ by Helix Development.